

ODD SEMESTER EXAMINATION- DECEMBER 2022

CLASS: B.TECH. SEMESTER: I PAPER

CODE: BEM-C102

PAPER TITLE: ENGINEERING MATHEMATICS

Time: 3 hour

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: Find the n^{th} derivative of $\frac{1}{(x+2)(x-1)}$.

Question-2: Expand $e^x \cos x$ by Maclacuin's theorem.

Question-3: Define Divergence and curl with their physical significance.

Question-4: Find the directional derivative of $f(x, y, z) = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of the vector $\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$.

Question-5: Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ by changing the order of integration.

Question-6: Discuss the maxima and minima of the following functions:

$$Z = x^2 + xy + 3x + 2y + 5$$

Question-7: Evaluate $\text{div } f$ and $\text{curl } f$ where $f = (xy \sin z)\mathbf{i} - (y^2 \sin x)\mathbf{j} + (yz^2 \sin x)\mathbf{k}$.

Question-8: A particle moves along the curve $x = t^3 + 1, y = t^2, z = 2t + 5$, where t is the time. Find the component of its velocity and acceleration at time $t=1$ in the direction $\hat{i} + \hat{j} + 3\hat{k}$.

Question-9: Employing elementary transformations, find the inverse of the matrix:

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

Question-10: Reduce the following matrix to normal form and hence find its rank:

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any **four** questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

Question-1: If $y = \sin^{-1}x$ prove that: $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$.

Question-2: If $u = \log(x^3+y^3+z^3-3xyz)$, show that:

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$$

Question-3: Evaluate $\int_S \hat{F} \cdot \hat{n} ds$ where $F = y^2z^2\hat{i} + z^2x^2\hat{j} + z^2y^2\hat{k}$ and S is the part of the sphere $x^2 + y^2 + z^2 = 1$ above the xy-plane bounded by xy-plane.

Question-4: If $x + y + z = u, y + z = uv, z = uvw$, show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$.

Question-5: If $F = y\hat{i} - x\hat{j}$, evaluate $\int_C F \cdot d\vec{r}$ from (0, 0) to (1, 1) along the following paths:

- The Parabola $y = x^2$
- The straight lines from (0, 0) to (1, 0) and then to (1, 1).

Question-6: Use Green's theorem to evaluate $\oint_C xydx + x^2y^3dy$ where C is the triangle with vertices (0,0), (1,0), (1,2) with positive orientation.

Question-7: Verify Cayley-Hamilton theorem for the matrix:

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & 2 \\ 5 & -5 & 11 \end{bmatrix}$$

Question-8: Test the consistency and hence solve the following set of equations:

$$x_1 + 2x_2 + x_3 = 2, \quad 3x_1 + x_2 - 2x_3 = 1, \quad 4x_1 - 3x_2 - x_3 = 3, \quad 2x_1 + 4x_2 + 2x_3 = 4.$$